

The Universal Hyperbolic Law: A Unified Framework for Quantum Yang-Mills Existence, Clifford Algebra, and the Observer Coupling Constant

Daniel J. Murray
Independent Researcher
thantiklermcirony@gmail.com

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Abstract

This paper unifies the geometric foundations of quantum mechanics and relativity by establishing the Universal Hyperbolic Law (UHL). We demonstrate that all bounded compositional systems are mathematically mandated to follow hyperbolic kinematics. We provide a rigorous construction for 3+1 Quantum Yang-Mills theory using a hyperbolic regulator, derive the Clifford Algebra from hyperbolic closure, and resolve the Measurement Problem via a universal observer-coupling constant, $\kappa \approx 10^{-6}$.

1 Introduction: The Boundedness Axiom

Physical quantities—from velocities capped at c to probabilities in $[0, 1]$ —are fundamentally bounded. This boundedness is not arbitrary but imposes a geometric structure on the dynamics of such systems. The Universal Hyperbolic Law (UHL) posits that any bounded, compositional one-dimensional quantity $u \in (-1/\kappa, 1/\kappa)$ must be linearizable by the rapidity coordinate $r = \frac{1}{\kappa}(\kappa u)$.

1.1 Establish that physical quantities are fundamentally bounded

In special relativity, velocities are bounded by the speed of light c , leading to the Einstein velocity addition law. In quantum mechanics, observables like momentum are constrained by physical limits, and probabilities are inherently bounded between 0 and 1. Similarly, in dynamical systems such as population genetics or measurement statistics, states evolve within fixed endpoints. The UHL generalizes this: boundedness implies hyperbolic geometry, where the bounding constant κ determines the curvature of the phase space.

1.2 Introduce the Rapidity Linearity Principle (RLP)

The Rapidity Linearity Principle (RLP) states that the inverse hyperbolic tangent $()$ is the unique linearizer for such systems. Under axioms of boundedness, associativity, and boundary-fixing Möbius translations, the composition law is uniquely $u \oplus v = \frac{u+v}{1+\kappa^2 uv}$. This derives from the requirement that left translations extend to real-analytic Möbius maps fixing the boundaries $\pm 1/\kappa$.

2 The Rigorous Hamiltonian Construction

We transpose the UHL to gauge theory by introducing a hyperbolic regulator to ensure boundedness and reflection positivity in lattice Yang-Mills (YM) theory.

2.1 Transpose the UHL to gauge theory using the tanh-regulator

The regulator is defined as $A_H = \kappa^{-1} \tanh(\kappa A)$, where A is the gauge field. This bounds the plaquette potential, ensuring gauge-invariance and reflection positivity (RP). The Hamiltonian is constructed on a lattice with kinematics, constraints, and Hilbert space defined to preserve these properties. The kinetic term is self-adjoint, and the potential is bounded with nonnegative character coefficients.

2.2 Cite the unconditional proof of the CK-2 Clustering Hypothesis

Using Doeblin-Harris minorization on the zero-momentum transfer operator, we prove LCP-uniform exponential clustering. The kernel is strictly positive and jointly continuous, yielding compactness and exponential mixing. This establishes an unconditional lattice-cutoff mass-gap lower bound uniform along lines of constant physics (LCP).

2.3 State the Conditional Mass Gap Theorem

If the bounded RP regulator and the Wilson action share the same Osterwalder-Schrader (OS) continuum limit along an LCP, then the continuum YM theory possesses a positive spectral gap: $\Delta_{YM} \geq \mu_* > 0$.

3 Geometric Derivation of Quantum Foundations

Applying the UHL to quantum phase space derives key structures geometrically.

3.1 Derive the Clifford Algebra (Dirac matrices)

The Clifford algebra emerges as the minimal structure required to satisfy the UHL's quadratic mass-shell invariant. Imposing linearity and the hyperbolic quadratic invariant on the first-order differential operator yields the anti-commutation relations $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. This is forced by the non-commutative composition in multi-dimensional gyrogroups, completing the transposition from bounded kinematics to relativistic quantum mechanics.

3.2 Introduce the Hyperbolic Canonical Commutation Relation (HCCR)

Assuming conjugate rapidity $\hat{r}$ obeys flat-space commutation, we derive the HCCR: $[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2)$. For $\kappa = 1/c$, this links special relativity's kinematic cap to quantum mechanics' observational bounds.

3.3 Present the Hyperbolic Uncertainty Principle (HUP)

The HUP follows: $\Delta x \Delta p \geq \frac{\hbar}{2}(1 - \kappa^2 \langle \hat{p}^2 \rangle)$, predicting vanishing quantum uncertainty near physical momentum bounds—a testable deviation from standard quantum theory. At the manifold level, monotone quantum information metrics pull back to hyperbolic form, yielding constant scalar curvature $R = -8\kappa^2$, resolving Petz's conjecture.

4 The Murray Reality Equation (MRE)

The MRE relates observer information processing to quantum measurement statistics, resolving the measurement problem.

4.1 Define the observer's information processing rate Φ

Φ is the "inverse pattern" that resolves superpositions into definite outcomes, quantified via EEG entropy rate or similar metrics. It represents the observer's capacity to process information, bounded and compositional under the UHL.

4.2 Present the MRE

The alternation bias δ (negative lag-1 autocorrelation) in binary quantum measurements is given by:

$$\delta(\Phi) = \tanh[(\delta_0) + 2\alpha_0\kappa\Phi]$$

where δ_0 is the unobserved baseline, α_0 is the device-specific transfer factor, and κ is the universal coupling constant. In the small-signal regime, $\Delta\delta \approx 2\alpha_0\kappa\Phi$. This preserves no-signaling and is mathematically necessary for bounded observer influence.

4.3 Synthesize the five independent derivations of κ

1. **Landauer's Thermodynamic Principle:** Coupling quantum action to information erasure yields $\kappa \sim 10^{-6}$ from thermodynamic limits. 2. **Bekenstein Holographic Bound:** Holographic deficit in bounded systems gives the same order. 3. **Kaniadakis Non-Extensive Statistics:** Non-extensive entropy for bounded observables converges to κ . 4. **Quantum Gravity Corrections:** Gravitational deformations in bounded manifolds predict κ . 5. **Feynman Path Integrals:** Path summation in hyperbolic space fixes $\kappa \approx 10^{-6}$.

All derivations converge, suggesting κ is a fundamental constant.

5 Discussion: The Loop of Self-Creation

5.1 Explain how the biophotonic scaffolding and limbic balance are the biological transpositions of the UHL

Biophotonic emissions in neural tissues provide the physical mechanism for Φ , with limbic balance ensuring monotonic information flow. This creates a closed loop: evolutionary processes select for efficient observers, whose statistics reflect hyperbolic structure.

5.2 Discuss the fractal limit set created by non-commuting observer drives

Non-commuting Möbius generators from multi-drive systems yield a fractal boundary limit set, encoding complexity in observer dynamics.

6 Conclusion and Falsifiability

The UHL provides a unified foundation for Information Geometry, bridging quantum foundations, relativity, and observer dynamics. It asserts hyperbolic kinematics as a mathematical necessity for bounded systems.

6.1 Assert that the UHL provides a unified foundation for Information Geometry

By proving uniqueness theorems and rigidity results, the UHL unifies domains from quantum field theory to biology under hyperbolic composition.

6.2 List the FLOW-GATE engineering falsifiers

1. **Small-signal linearity:** Test linear approximation $\Delta\delta \approx 2\alpha_0\kappa\Phi$ with Geiger-Müller detectors and EEG. 2. **Shuffle tests:** Permutation/bootstrap to falsify non-random alternation bias. 3. **Thomas Gyration detection:** Detect non-commutative gyration in multi-dimensional drives, confirming fractal limits.

These provide a definitive experimental path forward. If falsified, the UHL is refuted; if confirmed, it resolves longstanding problems in physics.

7 References

The references for this unified paper are primarily the component papers that form its basis, along with key sources cited within them. Below is a compiled list drawn from the individual documents:

1. Murray, D.J. (2025). Universal Hyperbolic Information Geometry.
2. Murray, D.J. (2025). The Hyperbolic Structure of Bounded Dynamical Systems.
3. Murray, D.J. (2025). Einstein Velocity Addition from Associativity and Boundary-Fixing Möbius Translations.
4. Murray, D.J. (2025). The Murray Reality Equation.
5. Murray, D.J. (2025). Unified Geometric Foundations: The Hyperbolic Canonical Commutation Relation.
6. Murray, D.J. (2025). Geometric Origin of Quantum Bounds.
7. Murray, D.J. (2025). The Dirac Transposition.
8. Murray, D.J. (2025). Yang-Mills in 3+1 Dimensions.
9. Murray, D.J. (2025). The Hyperbolic Dirac Equation.
10. Murray, D.J. (2025). Transpositions of a Universal Hyperbolic Law.

Additional key references from the component papers include:

- J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, 1966. (Cited in multiple papers for functional equation theory.)
- Dirac, P. A. M. (1981). *The Principles of Quantum Mechanics*. Oxford University Press. (Cited in quantum foundations.)
- Schlosshauer, M. (2019). Quantum decoherence. *Physics Reports*, 831, 1–57. (Cited in measurement problem discussions.)
- Gell-Mann, M., & Hartle, J. B. (1990). Quantum mechanics in the light of quantum cosmology. In *Complexity, Entropy and the Physics of Information*.
- Other sources such as works on gyrogroups by A. A. Ungar and quantum information metrics by D. Petz.

For complete bibliographies, refer to the References sections in the original PDFs, typically on the final pages.